

Emilio Izquierdo

Cleveland, OH

Phone: 716-468-6888 | Email: emilioizq27@gmail.com

EDUCATION

Case Western Reserve University, Cleveland, OH

August 2023 – May 2026

B.A. in Psychology (Honors) & Cognitive Science

Magna Cum Laude | GPA: 3.95 | Accelerated 3-year graduation

HONORS & AWARDS

Dean's High Honors, Case Western Reserve University

December 2023 – May 2026

National Hispanic Scholarship Fund Recipient

2023, 2024, 2025

HSF National Leadership Conference Delegate, Los Angeles, CA

August 2024

USA Gymnastics Academic All-American Award

2021 – 2023

SKILLS

- **Programming Languages:** Python (signal processing, data analysis), Java, R (data analysis and visualization), SQL
- **Data Management & Analysis:** REDCap (data collection, tracking, participant communication), Box (data collection, tracking), Excel, OpenRefine, JASP
- **Signal Processing:** electrocardiogram, impedance cardiography, and respiratory data cleaning and analysis
- **Languages:** Spanish

RESEARCH EXPERIENCE

Methodology & Research Collaborator

January 2025 – Present

In collaboration with Dr. Brooke Macnamara (Purdue University)

Mentored by Dr. Kyle LaFollette (University of Chicago)

Project Title: Psychophysiological Mechanisms of Stress and Cognitive Control

Description: Examines how acute stress affects cognitive control mechanisms by measuring physiological stress responses during the Maastricht Acute Stress Test (MAST). The project integrates custom signal processing pipeline development with empirical research examining

how individual differences in cardiovascular reactivity predict changes in attentional control processes assessed through Flanker task performance and diffusion decision modeling.

Signal Processing and Methods Development:

- Developed a comprehensive Python-based signal processing pipeline for extracting cardiovascular indices from impedance cardiography (ICG) data, integrating electrocardiography (ECG) and respiratory (RSP) signals simultaneously measured with ICG to derive hemodynamic markers including pre-ejection period (PEP), left ventricular ejection time (LVET), stroke volume (SV), velocity index (VI), and Heather index (HI)
- Solely responsible for designing and implementing the complete pipeline, including signal preprocessing (bandpass filtering, artifact correction, ensemble averaging), waveform feature extraction (B-point, C-point, X-point detection), and hemodynamic calculations using validated formulas from the impedance cardiography literature
- Integrated signal preprocessing techniques based on established methods, including Lipponen heartbeat correction algorithms for ECG artifact detection and adaptive filtering strategies to improve ICG point detection accuracy and waveform clarity
- Implemented the pipeline using NeuroKit2 and custom Python scripts, with core structure based on protocols from Karpiel et al. (2022) and additional methodological components synthesized from complementary literature
- Actively working toward validation and integration of the tool into open-source frameworks (e.g., NeuroKit2) to promote reproducibility in psychophysiological research

Empirical Research and Analysis:

- Extracted physiological indices from participant data collected during multi-phase experimental protocol (baseline, pre-stress Flanker task, MAST stress induction, post-stress Flanker task, recovery), calculating mean values, variance, and change scores for primary measures (VI, HI, PEP, MAP, HRV) and secondary measures (CO, SV, ACI, STR, respiratory variability)
- Conducted group-level analyses examining stress reactivity patterns across experimental phases and their relationships to cognitive control outcomes
- Co-authored preregistration examining how physiological stress responses predict both behavioral performance (reaction time effects, post-error slowing) and underlying cognitive mechanisms (response caution, distraction, processing speed) identified through computational modeling of decision-making (drift diffusion models)

Capstone Research Project

January 2026 – May 2026

Case Western Reserve University

Principal Investigator: Dr. Norah Feeny

Mentor: Isabelle Tully

Project Title: Emotion Dysregulation and Emotion Coping Outcomes After PTSD Treatment

Description: Independent capstone project using data from a randomized controlled trial comparing prolonged exposure (PE) to PE+Sertraline in adults with PTSD (N = 149).

- Developed research question in collaboration with mentor examining whether baseline emotion regulation difficulties relate to PTSD severity and maladaptive coping strategies at intake and following PE-based treatment
- Conducted between-groups analyses in R using Welch's independent samples t-tests, comparing high (DERS ≥ 95) and low baseline emotion regulation difficulties groups on PTSD severity, posttraumatic avoidance, anger suppression and expression, and dissociation
- Authored full capstone manuscript including introduction, methods, results, and discussion sections

PTSD Treatment and Research Center

June 2025 – May 2026

Case Western Reserve University

Research Assistant

Principal Investigator: Dr. Norah Feeny

Project Title: Positive Processes and Transition to Health (PATH)

Description: NIMH-funded clinical trial evaluating a novel treatment that targets maladaptive event processing in adults suffering from PTSD, depression, and trauma-related symptoms.

- Monitoring participant progress and creating alerts via REDCap
- Facilitating participant recruitment, administering scoring protocols, and organizing REDCap project configurations
- Facilitating journal article discussions during lab meetings

Research on Emotion and Attachment in Childhood Lab June 2025 – August 2025

University of Wisconsin–Madison

Summer Research Fellow

Program: Psychology Research Experience Program (PREP)

Principal Investigator: Dr. Kathryn Howell

Project Title: Pregnant Moms' Empowerment Program (PMEP)

Mentor: William Dworz

Description: A brief group intervention for pregnant women with histories of intimate partner violence, grounded in cognitive behavioral therapy, empowerment theory, and the social ecological theory of resilience. The program aims to improve prepartum and postpartum outcomes for both mothers and infants while connecting participants to community-based resources and services.

- Transcribed 4 hours of group intervention sessions (2 sessions, both 2 hours long) for qualitative analysis of participant experiences and therapeutic processes
- Identified and implemented efficiency improvements for transcription processes using available AI tools

Project Title: DAIS Focus Groups - IPV and Parenting Experiences

Mentor: William Dworz

Description: In partnership with Domestic Abuse Intervention Services (DAIS), conducted focus groups investigating experiences of parents who have experienced intimate partner violence, exploring their strengths, resilience, parenting relationships, and gathering feedback on support services.

- Reviewed IRB submission materials and provided feedback to ensure comprehensive protocol coverage
- Conducted phone-based participant eligibility screening for focus group recruitment
- Coordinated community outreach efforts, including flyer distribution and partnership development with local organizations
- Provided childcare support during focus group sessions to facilitate parent participation
- Tracked recruitment progress and maintained study documentation in Box

Project Title: Differential Prediction of PTSD Clusters by Emotion Regulation Difficulties in College Students

Mentor: William Dworz

Description: Conducted a secondary data analysis examining how specific emotion regulation difficulties differentially predict DSM-5 PTSD symptom clusters in a multi-site college sample. The study challenges traditional approaches that treat PTSD as a uniform construct by investigating which regulatory deficits predict which symptom patterns, with implications for targeted assessment and intervention approaches.

- Independently identified research question and selected appropriate dataset from available multi-site college student data
- Designed and conducted comprehensive statistical analysis plan including hierarchical regression with multiple robustness checks

- Independently created R script to implement advanced statistical techniques including robust standard errors, FDR correction, bootstrap confidence intervals, and collective significance testing
- Independently interpreted results and prepared findings for conference presentation

Project Title: Parent-Child Interventions for Youth Suicide Prevention (Collaborative Literature Review)

Mentors: Veronika Kobrinsky, William Dworz

Description: Collaborative systematic review with the Affect, Suicide, Self-Injury, and Social Triggers (ASSIST) Lab identifying evidence-based family- and caregiver-involved interventions for youth experiencing suicidality, trauma, or severe mental illness. Initial scope focused on youth under 18; refined scope targeted children ages 8-12 with emphasis on process-oriented intervention targets such as parent-child closeness and treatment engagement.

- As one of two primary reviewers, systematically reviewed and extracted data from 24 articles in initial search and 20 articles in refined search using standardized coding protocol
- Completed comprehensive data extraction for approximately half of all included studies, coding research objectives, sample characteristics, intervention descriptions, methodologies, outcomes, and limitations

PREP Program Components

- Selected for a competitive, fully funded summer research fellowship designed to prepare undergraduates for PhD training in psychology
- Received technical training in Python, R, and high-throughput computing for large-scale data analysis
- Presented research findings at the UW–Madison PREP Summer Research Symposium

Skill, Learning, and Performance Laboratory

September 2023 – May 2025

Case Western Reserve University (now at Purdue University)

Principal Investigator: Dr. Brooke Macnamara

Project Title: Affective States as Predictors of Learning and Performance

Description: Examined how individual affective traits (e.g., anxiety sensitivity, intolerance of uncertainty) and physiological stressors (e.g., cold pressor test, mental arithmetic) influence learning outcomes. Participants completed repeated trials of a game-like task while physiological signals (ECG, RSP, BP) were recorded. Findings contribute to understanding stress-performance interactions in cognitive training.

As Research Assistant (Sep 2023–Dec 2024):

- Ran two 4-hour experimental sessions per week
- Coordinated participant scheduling and executed data entry in REDCap
- Resolved technical issues collaboratively during experimental sessions

As Senior Research Assistant (Jan 2025–May 2025):

- Supervised a junior research assistant weekly, providing mentorship and oversight

Project Title: Dimensions of Difficulty in Video Game Play

Description: Study designed to identify cognitive and task-related features that influence difficulty in improving performance. Participants completed standardized cognitive tasks and an adaptive video game across two sessions. Findings aim to inform cognitive training and learning models in high-demand environments.

As Senior Research Assistant (Jan 2025–May 2025):

- Independently conducted two 2-hour cognitive sessions per week, ensuring experimental protocols were followed
- Managed participant scheduling and data entry in REDCap and Box

PRESENTATIONS

Izquierdo, E. R., Dwortz, W. R., Wamser, R. A., & Howell, K. H. (2025, August). Differential prediction of PTSD clusters by emotion regulation difficulties in college students [Conference presentation]. PREP Summer Research Symposium, University of Wisconsin–Madison, Madison, WI.

Izquierdo, E. R., Tully, I. A., & Feeny, N. C. (2026, April). Emotion dysregulation and emotion coping outcomes after PTSD treatment [Poster presentation]. Intersections Undergraduate Research Conference, Case Western Reserve University, Cleveland, OH.

VOLUNTEERING

University Hospitals Rainbow Babies & Children's Hospital January 2026 – May 2026

Cleveland, OH

Rainbow Unit Volunteer, Family & Child Life Services

Description: Volunteer with the Child Life team on an inpatient pediatric hematology/oncology unit, supporting immunocompromised children and their families through developmentally appropriate play and unit-based support.

- Support pediatric patients (infant–18 years) through age-appropriate play and one-on-one engagement in activity center and bedside settings

- Maintain a clean, safe, and non-threatening play environment in accordance with infection control protocols
- Collaborate with Child Life staff and nursing personnel to support patient comfort, transitions, and unit operations
- Adhere to patient confidentiality, safety guidelines, and hospital policies in a high-acuity inpatient setting

PLANNED RESEARCH & FUTURE WORK

Collaborative Research - University of Memphis

Summer 2026

Dissertation Collaborator: Hannah Gilliam

Project Title: Scoping Review - Parenting Among Mexican Mothers Experiencing IPV

Description: Collaborative systematic scoping review examining parenting practices and interventions among Mexican mothers experiencing intimate partner violence. Contributing to systematic coding and data extraction processes to identify patterns, gaps, and intervention targets in this understudied population.

- Systematic article screening and data extraction
- Cross-cultural analysis of parenting interventions and outcomes
- Synthesis of findings to inform culturally responsive intervention development

ADDITIONAL SKILLS

- Jazz drummer (professional performance experience)

CERTIFICATIONS

Group 2 Social & Behavioral Research Faculty, Staff, and Students

Expires September 2026

Mandated Reporter Training

Completed May 2025

GCP – Social and Behavioral Research Best Practices for Clinical Research Certification

Expires May 2028

UW Human Subjects Protections Course

Expires May 2028

CITI Health Information Privacy and Security for Clinical Investigators

Expires June 2028